## Team Details

a. **Team name:** **FairLens AI**

b. **Team leader name:** **Shreyash Pandey**

c. **Problem Statement:** **Unbiased AI Decision**

SOLUTION SUMMARY

# Brief About The Solution

CORE IDEA

## What FairLens AI does

Uploads or connects datasets, measures bias, explains risky decisions, and recommends fixes before deployment.

TARGET USERS

## Who uses it

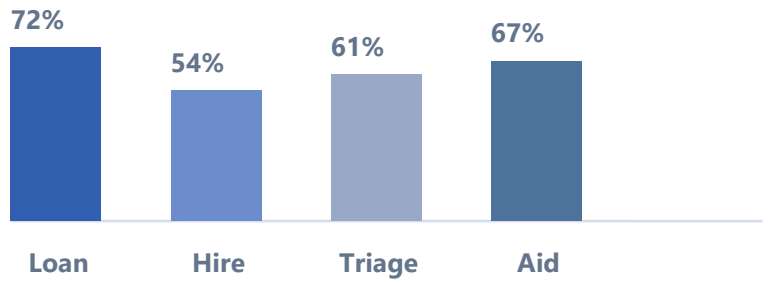
Compliance officers, data scientists, NGOs, schools, hospitals, and lending teams using automated decision systems.

GOOGLE AI

## Why Google AI is included

Gemini converts fairness metrics into plain-language audit explanations and safe remediation steps.

### Expected need across sectors




## Prototype Value Proposition

1. Detects unfair patterns early
2. Makes fairness understandable for non-technical reviewers
3. Gives a clear audit, explain, fix workflow
4. Supports cloud deployment and future reporting

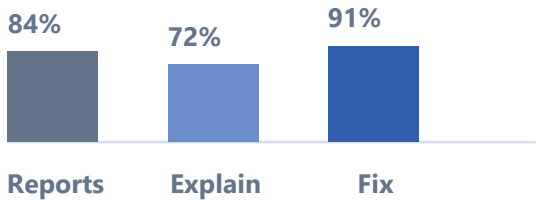
WHY THIS STANDS OUT

# Opportunity, Differentiation, And USP

## Different from existing ideas

Most bias tools stop at metrics. FairLens AI adds visual dashboards, plain-language explanations, and guided mitigation.

### Coverage of workflow




## How it solves the problem

Fairness tests identify disparity, Gemini surfaces root causes, and review steps prevent unsafe model release.

## USP of the proposed solution

One platform that combines fairness metrics, decision replay, AI explanation, remediation planning, and approval tracking.

### USP summary

One workflow for audit, explanation, mitigation, and approval tracking.

FEATURE MAP

# Features Offered By The Solution

SCAN

## Bias scan

Checks parity gaps across groups and flags imbalance quickly.

INSIGHT

## Gemini narratives

Explains fairness risks in a way judges and reviewers can understand.

TRACE

## Decision replay

Visualizes the path of one model decision and highlights risky inputs.

FIX

## Mitigation planner

Suggests thresholds, reweighting, and human review checkpoints.

REPORT

## Audit reports

Creates shareable fairness summaries with action items.

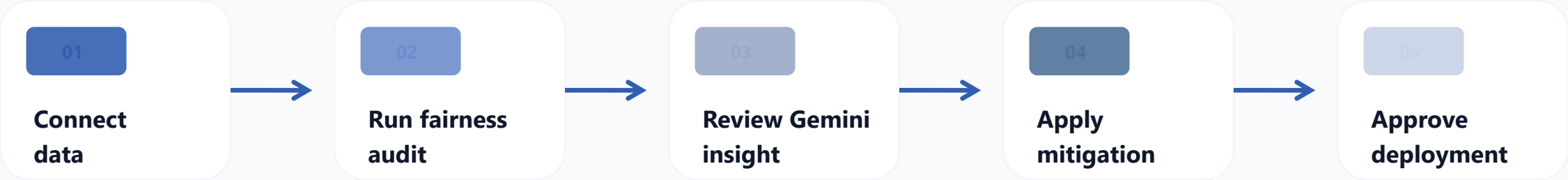
WATCH

## Live monitoring

Tracks post-deployment drift so fairness does not degrade silently.

END-TO-END USER JOURNEY

# Process Flow Diagram

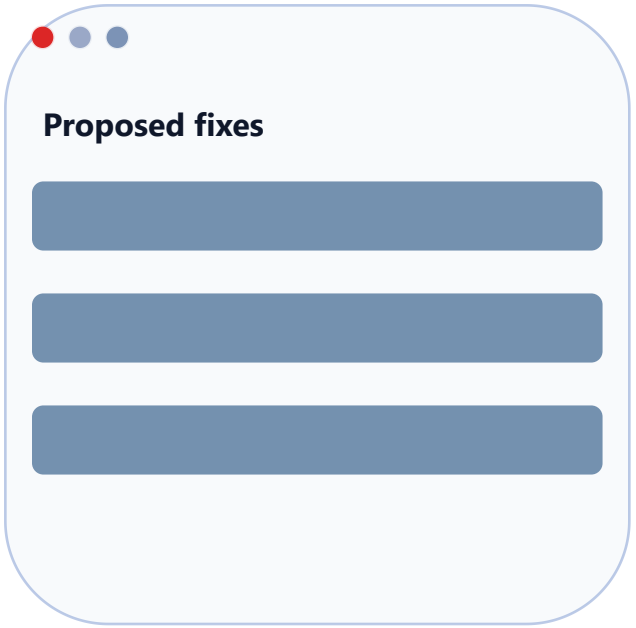


VISUAL PROTOTYPE

# Wireframes / Mock Diagrams

Admin fairness dashboard

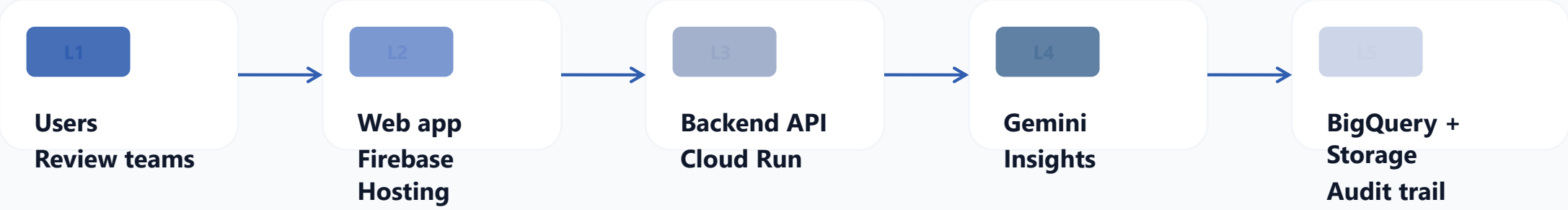
Decision replay panel




Mitigation sandbox

CLOUD + GOOGLE AI

# Architecture Diagram




## Why this architecture is strong

The user reviews results in the web app, Cloud Run handles processing, Gemini explains risky decisions, and BigQuery stores long-term audit history.

STACK OVERVIEW

# Technologies Used In The Solution

UI

## Frontend Layer

HTML, CSS, JavaScript, responsive dashboard design, and simple chart components for an accessible review experience.

INTELLIGENCE

## AI + Analytics Layer

Gemini API, fairness scoring logic, explainability prompts, bias-gap calculations, and remediation recommendations.

DEPLOYMENT

## Cloud Layer

Firebase Hosting, Cloud Run, BigQuery, Cloud Storage, and deployment-ready workflows for scaling the MVP.

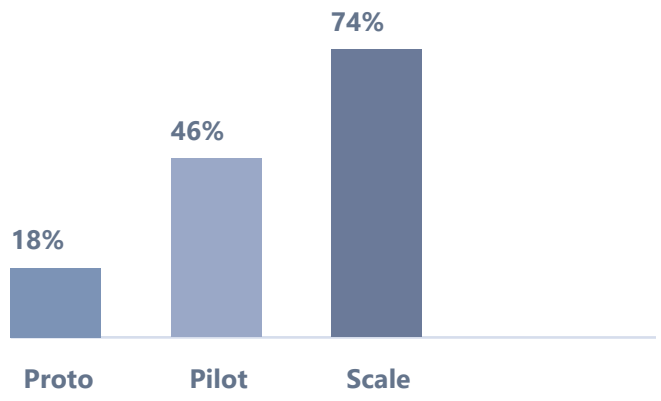
## Toolchain Fit

- Fast to prototype
- Easy to demo
- Cloud-native
- Judge-friendly
- Scalable later

COST OVERVIEW

# Estimated Implementation Cost

## Relative cost growth




LOW

### Prototype

Mostly free-tier usage with lightweight Gemini requests and static hosting.

MEDIUM

### Pilot

Moderate spend for inference, storage, and audit history as teams start testing.

HIGHER

### Scale

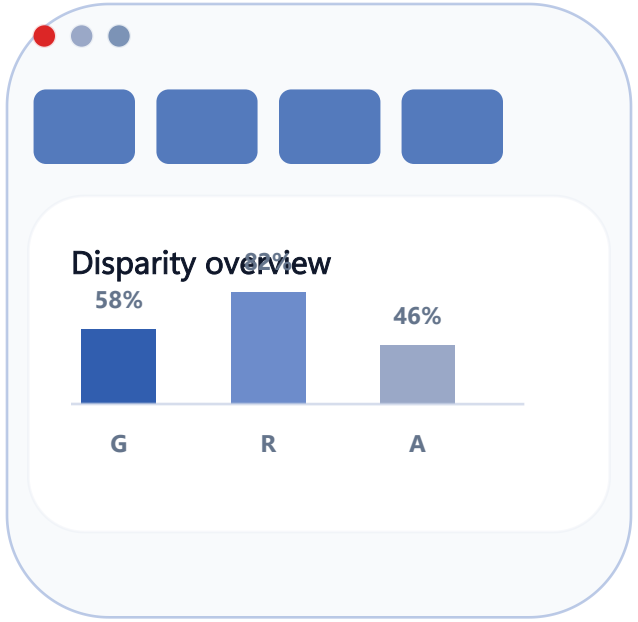
Higher cloud usage but recoverable through subscriptions for regulated sectors.

## Cost strategy

Start lean, validate with pilots, then scale into a subscription-based fairness auditing platform for institutions.

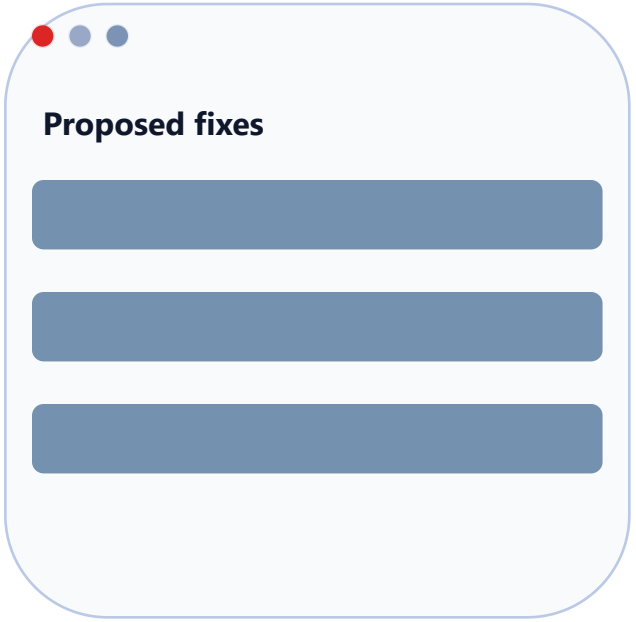
PRODUCT PREVIEW

# Snapshots Of The MVP




Audit dashboard

Gemini explanation




Mitigation planner

ROADMAP

# Additional Details / Future Development

01

## Sector templates

Hiring, lending, healthcare, and education specific audit templates.

02

## Policy packs

Localized fairness checks, documentation, and report formats.

03

## Live monitoring

Production drift alerts and recurring fairness checks after deployment.

04

## Approval workflow

Human sign-off, remediation history, and stronger audit trails.

## Future vision

The long-term product can become a fairness operating system for institutions that use AI in high-impact decisions.

FINAL CHECKLIST

# Submission Links

**GitHub Public Repository**

https://github.com/myself-archita/fairlens-ai

**Demo Video Link (3 minutes)**

Demo video link to be added

**MVP / Working Prototype Link**

https://fairlens-ai.vercel.app

WHY THIS IS SUBMISSION-READY

# Impact And Readiness

### Social impact

## High

Targets unfair outcomes in sectors that affect livelihoods and opportunities.

### Deployment readiness

## Cloud-ready

Designed for Firebase Hosting, Cloud Run, Gemini, and audit data storage.

### Judge clarity

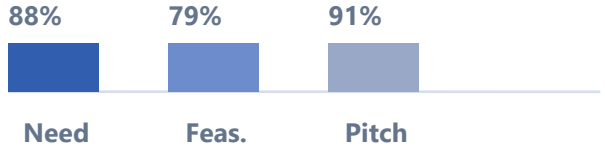
## Visual

Easy to explain with charts, flows, screenshots, and practical use cases.

## Why this prototype is strong

- **Solves a clear real-world problem**
- **Includes a Google AI model in the workflow**
- **Has a feasible deployment path, not just an abstract idea**

**Submission strength**



FAIRLENS AI

# Thank You

## Building fairer automated decisions with explainable AI.

FairLens AI combines bias detection, Gemini-powered explanation, remediation planning, and deployment-ready architecture in one student-friendly prototype.

Bias Detection

Gemini Insights

Cloud Ready